LOT (R) COMPUTER

FIRMWARE MANUAL -- INTERNAL / BUILD -- 3 July 2026

LOT SYSTEMS CORPORATION

LOT® COMPUTER — FIRMWARE SPECIFICATION

SEPARATE FROM HARDWARE SPEC AND SOFTWARE/API SPEC BY DESIGN

DOCUMENT FIRMWARE-SPEC / LOT-COMPUTER-v1

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CLASS INTERNAL / BUILD

RELATED docs/technical/LOT-COMPUTER-RIG-SPEC.md (hardware)

docs/technical/LOT-COMPUTER-SOFTWARE-BRIDGE.md (companion software / API)

Kept as an independent document because firmware ships on a slower, riskier cadence than platform software — an OTA firmware bug bricks physical units in the field; a bad platform deploy rolls back in seconds. The two must be versioned, reviewed, and released on separate clocks.

00 RUNTIME

TARGET ESP32-S3-WROOM-1 (N8R2)

FRAMEWORK ESP-IDF (not Arduino core) — direct control of deep-sleep states and the DVP camera interface matters more here than Arduino convenience.

LANGUAGE C, with a thin C++ layer for the session-compression buffer

01 STATE MACHINE

```
                ■■■■■■■■■■■■■■■
  power-on    ■   BOOT      ■
  ■■■■■■■■■■■■■ self-test   ■
                 ■■■■■■■■■■■■■■■
                       ■
                       ▼
                 ■■■■■■■■■■■■■■■
       ■■■■■■■■     IDLE     ■■■■■■■■
       ■     ■ e-paper held■     ■
       ■     ■ deep sleep  ■     ■
       ■     ■■■■■■■■■■■■■■■      ■
       ■           ■            ■
 display refreshed ■      Copy tap detected
 (message arrived) ■            ■
       ■           ▼            ■
       ■     ■■■■■■■■■■■■■■■      ■
       ■■■■■■■■    ACTIVE    ■■■■■■■■
             ■ radio wake  ■
             ■ sensor read ■
             ■■■■■■■■■■■■■■■
                   ■
                   ▼
             ■■■■■■■■■■■■■■■
             ■ SESSION     ■
             ■ COMPRESS    ■■■■■ see §03
             ■■■■■■■■■■■■■■■
                   ■
                   ▼
             ■■■■■■■■■■■■■■■
             ■ SYNC        ■■■■■ see software-bridge doc
             ■ (BLE/Wi-Fi) ■
             ■■■■■■■■■■■■■■■
```

The device spends its life in IDLE. E-paper holds the last message with zero standing current. ACTIVE is entered only by: (a) a message arriving over the radio, (b) a Copy tap on the piezo sensor, or (c) the periodic weather-sensor sample interval (default: every 15 minutes).

02 SENSOR DRIVERS

Sensor	Interface	Sample Rate	Notes
BME680 (weather/gas)	I2C	1 sample / 15 min	Bosch BSEC library for gas-baseline compensation
OV2640 (camera)	DVP (native ESP32-S3)	on-demand only	Never free-running. Captures a single low-res frame for ambient-ligh
Piezo disc (Copy button)	ADC, threshold + debounce	interrupt-driven	Same GPIO also drives haptic tick on confirmation (bi
BQ51013B (Qi)	I2C (status only)	polled on wake	Charge state feeds power-budget logic, not user-facing telemetry

Camera use is intentionally narrow. The OV2640 is a presence/ambient-light sensor with a lens, not a recording device. Firmware never buffers more than one frame, never writes a frame to flash, and the frame itself never crosses the radio — only a derived scalar (**ambient_lux**, **presence: bool**) leaves the device. This is a hard firmware invariant, not a policy promise: there is no code path that serializes raw frame bytes to the sync buffer.

03 SESSION COMPRESSION — ON-DEVICE

Mirrors the compression discipline already documented for the platform's Memory Engine (**docs/technical/MEMORY-ENGINE-COMPRESSION-ARCHITECTURE.md**): raw signal in, one dense fact out. A "session" here is bounded by IDLE→ACTIVE→IDLE.

```
SESSION COMPRESSION BUFFER (RAM, cleared every sync)
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
raw samples over the session window:
  - N x BME680 readings (temp, humidity, pressure, gas)
  - camera-derived ambient_lux, presence bool (never raw frames)
  - copy_tap: bool, tap_timestamp
  - battery_mv, charge_state
        ■
        ▼  on-device reduction (mean/median/delta, not raw log)
        ▼
ONE COMPRESSED RECORD PER SESSION:
{
  deviceId, sessionStart, sessionEnd,
  weather: { tempC, humidityPct, pressureHpa, vocIndex },  // reduced, not raw series
  presence: { seen: bool, ambientLux },
  copy: { tapped: bool, tapTimestamp },
  battery: { mv, chargeState }
}
```

The device never accumulates history across sessions. Once a compressed record is synced (software-bridge doc), the RAM buffer is cleared. If sync fails, exactly one record queues for retry — the device does not become a local data store. This keeps the firmware's failure mode simple: worst case is a dropped notification, never a privacy incident from an accumulating on-device log.

04 DISPLAY UPDATE RULE

Firmware enforces the Ambient AI™ "one line, no alarm" rule at the lowest level, not just as a design guideline:

- A new message triggers **partial refresh only** (~0.3s, no flash) unless 8+ hours have elapsed since the last full refresh (e-paper ghosting correction — a full refresh is scheduled automatically, never on demand).

- Maximum message length is enforced in firmware (not just by the sender): one line, ~24 characters at the display's default type size. A longer string sent to the device is truncated firmware-side, not word-wrapped — this keeps the hardware honest to the brand rule even if a future software bug tries to send more.
- No blink, no inverted flash-alert state. A haptic tick (piezo disc, §02) is the only non-visual acknowledgment, fired once on message arrival.

05 POWER BUDGET

STATE	CURRENT DRAW	DUTY
Deep sleep / IDLE	<25uA	~99% of lifetime
BME680 sample	~1.5mA for ~50ms	every 15 min
Wi-Fi/BLE sync burst	~120mA for ~2-4s	per session, on Copy tap or push
E-paper refresh	~15mA for ~0.3-2s	per message
Camera single frame	~80mA for ~100ms	on-demand only

Given the bridge-cell battery decision in the rig spec (§07 there), firmware defaults to Wi-Fi OFF / BLE-only sync when the device has been off the charging puck for >30 minutes, falling back to Wi-Fi direct only if BLE companion sync (software-bridge doc) is unavailable. This is a firmware-level power policy, tunable but shipped with a conservative default for v1.

06 OTA UPDATE

- Dual-partition (A/B) OTA via ESP-IDF's native **esp_ota** — a failed update rolls back to the last-known-good partition automatically on watchdog timeout. No bricking on a bad flash.
- OTA payloads are signed; the bootloader rejects unsigned images. Given 100 units in the field talking to a production API, an unsigned-OTA path is not acceptable even at pilot scale.
- OTA is pulled by the device on its own schedule (checked once per day during a sync burst), never pushed unsolicited — consistent with the "no silent writes" transparency principle already documented for LOT® server infrastructure (**docs/technical/LOT-NODE-0-RIG-SPEC.md**, §04).

07 WHAT FIRMWARE EXPLICITLY DOES NOT DO

- Does not store raw camera frames, ever, in flash or RAM beyond one frame.
- Does not accumulate a local log across sessions (see §03).
- Does not accept unsigned OTA images.
- Does not render more than one line of text regardless of what is sent.
- Does not attempt local Wi-Fi provisioning UI on a 200x200 e-paper screen — provisioning is BLE-only, driven from the companion app (software-bridge doc §01), because there is no keyboard on this device and there will never be one.

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END OF SPECIFICATION 2026.07.03